Basic Buck-Boost Converter — Step-Up/Step-Down DC–DC Conversion

Mini Paper — Power Electronics Fundamentals

Objective

This note examines the open-loop behavior of a non-isolated buck-boost converter using PSIM simulation.
The focus is on:

- The relationship between duty cycle and output voltage magnitude.
- The effect of inductance and capacitance on transient response in both step-down and step-up regions.

The goal is to observe how identical passive components behave under both operating modes before introducing closed-loop control.

System Model

Input voltage: 25 V DC
Load: 5 Ω resistive
Control method: Open-loop PWM
Duty ratios evaluated: D = 0.3 (buck region), D = 0.7 (boost region)
Switching frequency: 20 kHz

All simulations operated in continuous conduction mode (CCM).

Operating Principle

When the MOSFET switch is **ON**, current flows from the input through the inductor, storing energy in its magnetic field:

$$VL = Vin$$

When the switch turns **OFF**, the inductor reverses polarity and releases energy through the diode into the output capacitor and load:

$$VL = Vin + Vout$$

The steady-state voltage relationship is given by:

$$D$$

$$v_0 = -\frac{D}{1-D} v_{in}$$

where is the duty cycle.

The negative sign represents inversion of output polarity relative to the input ground.

Circuit Parameters

PARAMETER	SYMBOL	VALUE	DESCRIPTION
INPUT VOLTAGE	(Vin)	25 V	Constant DC source
LOAD RESISTANCE	(R)	5 Ω	Represents load
INDUCTANCE	(L)	5mH	Controls current ramp
CAPACITANCE	(C)	200 µF	Smooths voltage ripple
SWITCHING FREQUENCY	(fs)	20 kHz	Determines ripple and response
DUTY CYCLE	(D)	0.3 / 0.7	Buck and boost test points

These parameters ensured continuous conduction and allowed both buck and boost behavior to be observed under identical hardware conditions.

Circuit Diagram

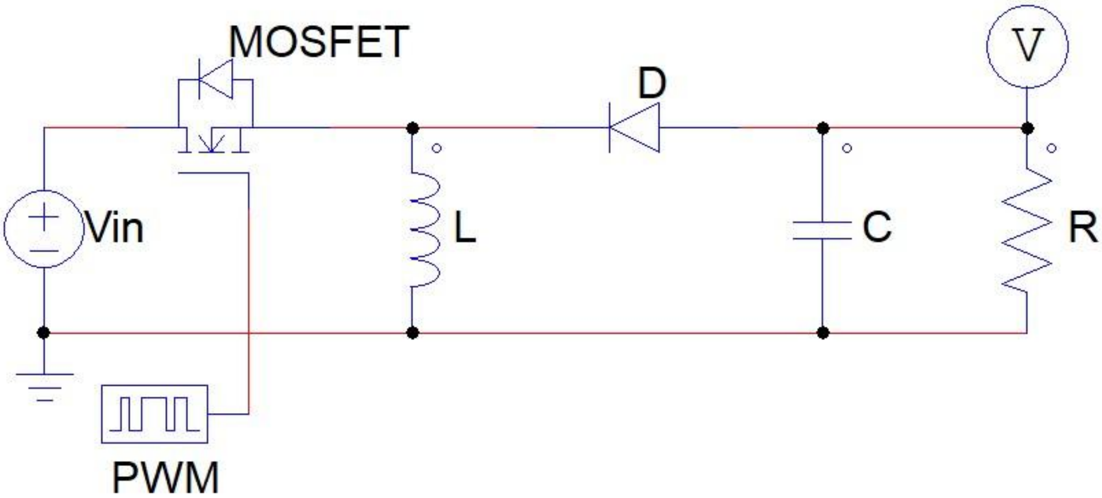

Figure 1

Simulation Observations

(a) Step-Down Region (D = 0.3)

• Output stabilized near −11 V, consistent with theory.
• A short startup overshoot was observed before settling.
• Average inductor current was approximately 3 A.

The overshoot is attributed to stored magnetic energy continuing to charge the output capacitor during the initial transient.

(b) Step-Up Region (D = 0.7)

• Output reached approximately −58 V, consistent with theoretical prediction.
• The voltage rise was smooth and monotonic.
• Average inductor current increased significantly (approximately 38 A).

Higher duty operation required substantially larger inductor current to maintain power balance between input and output.

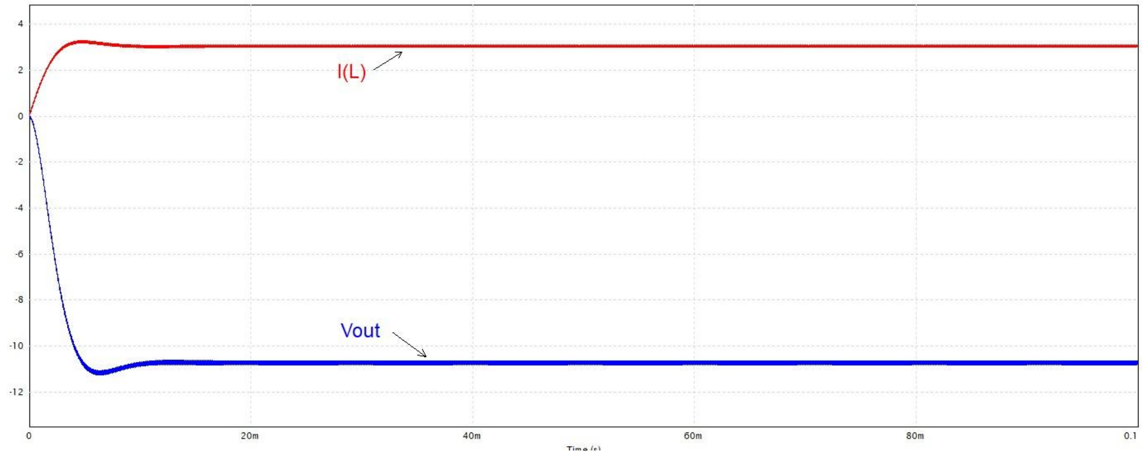

Figure 2 Step-Down Operation

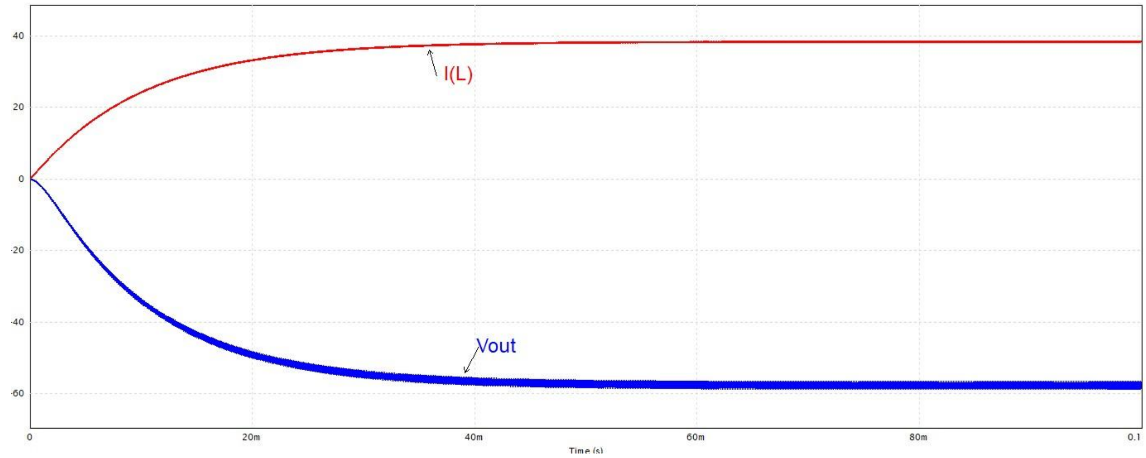

Figure 3 Step-Up Operation

Parameter Sensitivity

Duty Cycle

• Determines both magnitude and direction of voltage conversion.
• For D < 0.5, |Vout| < Vin (buck region).
• For D > 0.5, |Vout| > Vin (boost region).
• Current stress increases rapidly at high duty ratios.

Inductance

• Larger L reduces current ripple in both modes.
• Large L improves damping in boost region but can increase transient overshoot in buck region depending on energy release dynamics.

Capacitance

• Larger C reduces output-voltage ripple.
• Excessive capacitance may prolong settling time during startup.

Summary

• The buck-boost converter provides both step-down and step-up conversion with inverted polarity.

• Output magnitude follows :

$$v_0 = -\frac{D}{1-D} v_{in}$$

under CCM.

• The same passive network exhibits different transient characteristics in buck and boost regions.

• High duty operation significantly increases inductor current stress.

• A single passive design cannot fully optimize both operating modes in open-loop operation.

This establishes the open-loop buck-boost behavior baseline before implementing voltage feedback and soft-start control.